



DEPARTMENT OF MATHEMATICS
Academic year 2024-2025 (Odd Sem)
(CIE II- B. E. PROGRAMS, EC, EE, EI, ET)

Date	10-01- 2025	Maximum Marks	10 +50
Time	09:30 AM to 11 :30 AM	Duration	120 Mins
Semester	III Sem B. E. (EC, EE, EI, ET)		
MA231TA - LINEAR ALGEBRA, FOURIER TRANSFORMS AND STATISTICS			

Q. No.	PART - A	M	BT	CO
1	Consider the vectors $v_1 = (1, 2, 3)$ and $v_2 = (2, 3, 1)$. Find k so that vector $u = (1, k, 4)$ is linear combination of v_1 and v_2 .	2	1	2
2	Find the values of k for which the following set of vectors $S = \left\{ \begin{bmatrix} 1 \\ k \end{bmatrix}, \begin{bmatrix} k \\ k+2 \end{bmatrix} \right\}$ is linearly dependent.	2	1	1
3	Find the matrix of linear transformation which is obtained by first rotating all vectors through an angle $\frac{\pi}{6}$.	2	2	2
4	The projection of $(1, 2, 3)^T$ onto the vector $(1, 1, 1)^T$ is _____.	2	1	2
5	If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$, the sum of eigen values _____ and product of eigen values _____.	2	1	1

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	4	6	-	-	8	2		-	-	-

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Q. No.	PART - B	M	BT	CO
1	Find basis and dimension of the four fundamental subspaces for the matrix $A = \begin{bmatrix} -3 & 6 & -1 & 1 & -7 \\ 1 & -2 & 2 & 3 & -1 \\ 2 & -4 & 5 & 8 & -4 \end{bmatrix}.$	10	3	2
2a	Determine whether the vectors $f(x) = 2x^3 + x^2 + x + 1$, $g(x) = x^3 + 3x^2 + x - 2$, $h(x) = x^3 + 2x^2 - x + 3$ in vector space $P_3(x)$, the set of polynomials of degree ≤ 3 of polynomials over the field $\mathbb{R}$ are linearly independent or not.	6	3	1
2b	Let $T: R^3 \rightarrow R^3$ be a linear transformation such that $T(-1,0,1) = (1,0,0)$, $T(0,1,-1) = (0,1,0)$, $T(1,-1,1) = (0,0,1)$. Find explicit formula for $T(x,y,z)$.	4	2	2
3a	Diagonalize the matrix $A = \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ -1 & 2 & 5 \end{bmatrix}.$	10	3	3
4a	Find the least squares solution of $Ax = b$: where $A = \begin{bmatrix} 2 & 0 \\ -1 & 1 \\ 0 & 2 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}.$	6	3	3
4b	Is the vector $v = (3, -1, 0, -1)$ in subspace of R^4 spanned by the vectors $v_1 = (2, -1, 3, 2)$, $v_2 = (-1, 1, 1, -3)$, $v_3 = (1, 1, 9, -5)$?	4	3	3
5	Determine QR factorization of the matrix $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$	10	2	2

BT-Blooms Taxonomy, CO-Course Outcomes, M-Marks

Marks Distribution	Particulars		CO1	CO2	CO3	CO4	L1	L2	L3	L4	L5	L6
	Test	Max Marks	06	24	20	-	-	14	36	-	-	-
